

1. Analyze the convergence of the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{3n+2}$

2. Analyze the convergence of the series $\sum_{n=1}^{\infty} \frac{n}{3^n}$.

3. Find the radius and interval of convergence of the power series $\sum_{n=1}^{\infty} \frac{2^n}{n} x^n$. Use that

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{2^{n+1}}{n+1} x^{n+1}}{\frac{2^n}{n} x^n} \right| = 2|x|.$$

4. Using a series you already know, find the Taylor series for $f(x) = xe^{4x}$.